Abolfazl Sajadi

Digital Hardware Engineer · Hardware Security

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Profile

Digital hardware engineer completing a PhD at Leiden University, with hands-on experience in RTL/SoC design, RISC-V integration, FPGA prototyping, ASIC implementation, and hardware security. Experienced in taking digital systems from RTL and verification through physical implementation and tape-out.

Experience

Aug 2022 – Feb 2027 · ONGOING

Doctoral Researcher — Digital IC & Hardware Security — Leiden University, LIACS · Leiden, NL

- Designed the peripheral RTL — SPI, configurable UART, control/status registers and a PRNG, and integrated the **PROACT dual-core RISC-V cryptographic SoC** around its **AES, Ascon and Xoodyak** co-processors, from RTL to full chip.
- Prototyped and validated the SoC on Xilinx Artix-7 and Zynq-7000 FPGAs using SPI/UART interfaces; developed C and Python libraries, a control GUI, and user documentation.
- Synthesized the SoC for GlobalFoundries 22FDX using Cadence Genus and performed functional and SDF-annotated gate-level verification in Xcelium.
- Contributed to ASIC physical implementation and MPW tape-out with the imec team, including place-and-route, sign-off, GDSII generation, and design of the 28-pin DIP package bonding diagram.
- Built a **ChipWhisperer CW308** target board (Altium) and automated **CPA/TVLA** side-channel evaluation of AES on the RISC-V core.

2018 – 2021

Research Assistant — Embedded Systems (full-time) — University of Tehran, DVD-ES Lab · Tehran, IR

- Designed and evaluated **ML-resistant PUF authentication** protocols (DC-PUF, SQ-PUF) on FPGA; implemented CNN/SVM modeling attacks in Python (published in *J. Network & Computer Applications*).

Selected Projects

Open-source

ASSESS — pre-silicon side-channel leakage analysis — ACM CF'26

- A cycle-accurate methodology that localizes **gate-level side-channel leakage** from toggle activity, fast enough for routine checks during iterative **ASIC design**; benchmarked against RTL-PAT, PATCH and ACA on RISC-V/AES designs.

Vivado HLS

Real-time object detection & tracking on Zynq-7000 — B.Sc. thesis

- HW/SW co-design on an ARM+FPGA SoC: HDMI video in, filtering/tracking synthesized via **High-Level Synthesis**, output over VGA/UART.

Cyclone II

CNN accelerator on FPGA — RTL / Quartus

- RTL implementation of a tiny convolutional-network accelerator (MNIST), validated against a Keras reference model and brought up on an Altera Cyclone II board.

ESP32 · AVR

Home meter monitoring device — Altium · 2020

- Embedded design on ESP32 and AVR (ATmega8) microcontrollers, with a custom PCB designed in Altium Designer.

Education

Aug 2022 – Feb 2027 · ONGOING

Ph.D. in Computer Science — Leiden University (LIACS) · Leiden, NL

Hardware security & digital IC design (PROACT). Dissertation: pre/post-silicon side-channel leakage evaluation in lightweight cryptographic circuits.

2018 – 2021

M.Sc. in Electrical Engineering — Digital Electronic Systems — University of Tehran · GPA 18.28/20

Ranked 1st in class. Thesis: ML-based modeling attacks on Physical Unclonable Functions (PUFs).

2014 – 2017

B.Sc. in Electrical Engineering — Electronic Technology — Mohajer Technical University (MTU) · GPA 16.87/20

Top of class. Thesis: real-time object detection & tracking on a Zynq-7000 SoC.

Selected Publications

- **A. Sajadi**, N. Zidarič, T. Stefanov, N. Mentens. *Systematic Comparison and Improvement of Pre-silicon Leakage Analysis Tools.* ACM Computing Frontiers (CF Companion '26), 2026. [doi]
- **A. Sajadi**, N. Zidarič, T. Stefanov, N. Mentens. *A Systematic Comparison of Side-channel Countermeasures for RISC-V-based SoCs.* IEEE NorCAS, 2024. [doi]
- **A. Sajadi**, A. Shabani, B. Alizadeh. *DC-PUF: ML-Resistant PUF-Based Authentication for Resource-Constrained IoT.* J. Network & Computer Applications, vol. 217, 2023. [doi]
- A. Adhikary *et al.* (incl. **A. Sajadi**). *PROACT — Physical Attack Resistance of Cryptographic Algorithms and Circuits with Reduced Time to Market.* Applied Reconfigurable Computing (ARC), LNCS, Springer, 2024. [doi]
- J. Anders *et al.* (incl. **A. Sajadi**). *A Survey of Developments in Testability, Safety and Security of RISC-V Processors.* IEEE ETS, 2023. [doi]

Technical Skills

Hardware Languages	VHDL, Verilog, SystemVerilog, High-Level Synthesis (HLS)
Programming	Python, C/C++, Tcl (EDA scripting)
ASIC Front-End	RTL design, RISC-V (Ibex) integration, logic synthesis, SDC constraints, STA, gate-level simulation
Back-End	Place-and-route, sign-off, GDSII, bonding diagram, (GF 22FDX)
Verification	RTL testbenches, functional simulation, SDF-annotated gate-level simulation, FPGA validation, hardware debugging
FPGA & Embedded	Xilinx FPGAs, Vivado & Vivado HLS, Catapult HLS, SPI/UART, ESP32, AVR
Hardware Security	CPA, DPA, TVLA, SNR/NICV, ChipWhisperer, pre-silicon leakage analysis, PUFs
Design Methods	HW/SW co-design, continuous integration (CI), test & testability design (DfT), fault-tolerant system design
ML & EDA	Cadence (Genus, Joules, Xcelium, ...), Altium, MATLAB, PyTorch, Keras

Teaching, Supervision & Activities

- **Teaching Assistant**, *System & Software Security*, Leiden University (2022–present); **co-supervised 6+ B.Sc. theses** on AES/RISC-V side-channel analysis and chip-prototype leakage assessment.
- **Reviewer**, *IEEE Trans. Computer-Aided Design (TCAD)*;
- **Cadence certified** — RTL-to-GDSII Flow, Genus Physical Synthesis, Tcl for EDA (2024–25).
- **Summer schools**: CPS Summer School 2023 (Sardinia); Real-World Crypto & Privacy 2023; PROACT Hardware-Security Training Schools (2023, 2025).

Honors & Languages

- **Honors**: Ranked 102nd of 30,000+ in Iran's national M.Sc. entrance exam; 1st place, provincial WorldSkills electronics competition (2013).
- **Languages**: Persian (native) · English (professional working) · Dutch (A2).